GPU-accelerated Bayesian inference of multi-species biofilm interaction parameters via TMCMC

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Abstract

We present a GPU-accelerated Bayesian framework for inferring the interaction parameters of a five-species oral biofilm model derived from the extended Hamilton principle. The symmetric 5×5 interaction matrix is parametrised as a 20-dimensional vector encoding 15 independent interaction coefficients plus 5 intrinsic growth rates. A two-phase estimation strategy is employed: in Phase 1, the viability fraction ψ_i is fixed to experimentally measured membrane-intact ratios (fix-ψ), enabling rapid identification of the interaction matrix; in Phase 2, ψ_i is released and the viability data enter the likelihood as a soft constraint, yielding a self-consistent posterior over both interactions and viability. The forward model is compiled via JAX and evaluated in parallel on GPU (NVIDIA RTX 4090), achieving a $\sim$200$\times$ speedup over the CPU baseline. No interaction parameters are locked to zero a priori; the posterior self-organises, concentrating pathogen-related entries near zero in commensal conditions and activating cooperative cross-feeding terms in dysbiotic states. The method is validated against in-vitro time-course data from Heine et al. under four combinations of community state (commensal/dysbiotic) and cultivation mode (static/HOBIC reactor).

Keywords: Bayesian inference, biofilm dynamics, TMCMC, GPU acceleration, parameter identification, multi-species interaction, peri-implantitis

1 Introduction

Multi-species biofilms on dental implant surfaces are a primary cause of peri-implantitis [1, 2], and recent in-vivo characterisation confirms that these communities are complex, patient-specific, and temporally dynamic [3, 4]. Heine et al. [5] characterised the in-vitro dynamics of five key oral species—*Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei), *Fusobacterium nucleatum* (Fn), and *Porphyromonas gingivalis* (Pg)—under four combinations of community state (commensal/dysbiotic) and cultivation mode (static/HOBIC reactor). Building on the extended Hamilton principle [6], Klempt et al. [7] formulated a continuum model in which a symmetric interaction matrix $\mathbf{A}$ governs species growth, with an optional decay vector $\mathbf{b}$ for antibiotic effects. Fritsch et al. [8] subsequently introduced Bayesian updating with surrogate models for biofilm parameter estimation under hybrid uncertainties.

Despite these advances, estimating the 20-dimensional parameter vector (15 interaction coefficients plus 5 growth rates) from limited time-course data remains challenging. Two principal difficulties arise: (i) poor identifiability when data are sparse relative to the parameter dimension—with $N_t = 5$ usable time points (Day 1 is consumed as initial condition) and 5 species, one has at most 25 observations for 20 unknowns—and (ii) multimodal posterior landscapes caused by the coupled ϕ_i–ψ_i dynamics, which render standard MCMC methods inefficient [9].

In this work we address these challenges through three complementary strategies. First, we fix the viability fraction ψ_i to experimentally measured membrane-intact ratios, eliminating ψ-related posterior modes and halving the ODE system dimension. Second, we compile the forward model in JAX [10] and evaluate the entire particle ensemble on GPU via batched `vmap`, achieving a $\sim$200$\times$ speedup over the CPU baseline. Third, we employ iteratively narrowed priors that concentrate sampling on the high-posterior-density region, enabling convergence in 5–7 TMCMC [11, 12] stages even in 20 dimensions. The framework is validated against all four experimental conditions of Heine et al., demonstrating quantitative reproduction of commensal homeostasis as well as the non-linear pathogen surge characteristic of dysbiotic biofilms under HOBIC flow.

2 Mathematical model

2.1 State variables and kinematics

Following Klempt et al. [7, 13], the biofilm is described at a material point by two sets of internal variables: the volume fraction $\phi_i \in [0,1]$ occupied by species i ($i = 1, \ldots, n$; here $n = 5$), and the viability fraction $\psi_i \in [0,1]$ representing the proportion of living cells within species i. The living volume fraction is then

$$\bar{\phi}_i := \phi_i \, \psi_i, \qquad i = 1, \ldots, n. \tag{1}$$

An additional void fraction ϕ_0 closes the system via the holonomic constraint

$$\sum_{l=0}^{n} \phi_l(t) = 1 \quad \forall \, t. \tag{2}$$

2.2 Variational formulation

The evolution equations are derived from the extended Hamilton principle [6, 14]

$$\mathcal{H} = \int_{\mathcal{T}} (G + C + D) \, \mathrm{d}t \to \mathrm{stat}_{\boldsymbol{\xi},\gamma} \,, \tag{3}$$

comprising the Helmholtz free energy G, the constraint functional C, and the dissipation D. The free energy density encodes nutrient-driven growth and antibiotic-induced killing:

$$\Psi = -\tfrac{1}{2} \, c^* \, \bar{\boldsymbol{\phi}}^T \mathbf{A} \, \bar{\boldsymbol{\phi}} + \tfrac{1}{2} \, \alpha^* \, \boldsymbol{\psi}^T \mathbf{B} \, \boldsymbol{\psi}, \tag{4}$$

where c^* is the nutrient concentration, α^* the antibiotic concentration, $\mathbf{A} \in \mathbb{R}^{n \times n}$ the symmetric growth coefficient matrix, and $\mathbf{B} = \mathrm{diag}(b_1, \ldots, b_n)$ the antibiotic sensitivity matrix. The volume constraint (2) enters through a Lagrange multiplier γ:

$$C = \gamma \Big(\sum_{l=0}^{n} \phi_l - 1 \Big). \tag{5}$$

The dissipation functional $D = \int \Delta_s \, \mathrm{d}V$ is derived from the rate-dependent potential [7]

$$\Delta_s = \tfrac{1}{2} \, \dot{\bar{\boldsymbol{\phi}}}^T \boldsymbol{\eta} \, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2} \, \dot{\boldsymbol{\phi}}^T \boldsymbol{\eta} \, \dot{\boldsymbol{\phi}}, \tag{6}$$

with the diagonal viscosity matrix $\boldsymbol{\eta} = \mathrm{diag}(\eta_1, \ldots, \eta_n)$. The first term couples $\dot{\phi}_i$ and $\dot{\psi}_i$ through $\dot{\bar{\phi}}_i = \dot{\phi}_i\psi_i + \phi_i\dot{\psi}_i$, giving rise to non-linear dissipative interactions; the second term regularises the system to full rank. A logarithmic barrier with penalty $K_p = \eta_i \times 10^{-4}$ enforces the box constraints $\phi_i, \psi_i \in (0, 1)$ [7].

2.3 Governing equations

Evaluating $\delta\mathcal{H} = 0$ with respect to variations $\delta\phi_i$, $\delta\psi_i$, and $\delta\gamma$ yields the coupled evolution equations [7]:

Volume fraction $(i = 1, \ldots, n)$:

$$0 = -c^*\,\psi_i\,(\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i\big(\dot{\phi}_i\,\psi_i^2 + \bar{\phi}_i\,\dot{\psi}_i + \dot{\phi}_i\big) + \gamma + \frac{\partial\Psi_{\mathrm{bar}}}{\partial\phi_i}, \tag{7}$$

Viability fraction $(i = 1, \ldots, n)$:

$$0 = -c^*\,\phi_i\,(\mathbf{A}\bar{\boldsymbol{\phi}})_i + \alpha^*\,b_i\,\psi_i + \eta_i\big(\dot{\psi}_i\,\phi_i^2 + \bar{\phi}_i\,\dot{\phi}_i\big) + \frac{\partial\Psi_{\mathrm{bar}}}{\partial\psi_i}, \tag{8}$$

Void fraction:

$$0 = \gamma + \dot{\phi}_0 + \frac{\partial\Psi_{\mathrm{bar}}}{\partial\phi_0}, \tag{9}$$

Volume constraint:

$$0 = \sum_{l=0}^{n}\phi_l - 1, \tag{10}$$

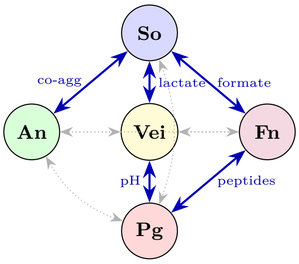
where $\Psi_{\mathrm{bar}}(\phi_i) = K_p(2-4\phi_i)/[(\phi_i-1)^3\phi_i^3]$ is the barrier penalty derivative. The system (7)–(10) constitutes $2n+2$ coupled non-linear equations for the unknowns $(\phi_1, \ldots, \phi_n, \phi_0, \psi_1, \ldots, \psi_n, \gamma)$, solved at each time step via Newton's method with an implicit Euler discretisation ($\Delta t = 10^{-4}$).

The variational derivation automatically ensures symmetry $A_{ij} = A_{ji}$ and thermodynamic consistency [6]. Since c^* enters the evolution equations only through the product c^*A_{ij}, its absolute value is non-identifiable; we fix $c^* = 25$ so that the inferred A_{ij} remain $\mathcal{O}(1)$. The Heine et al. experiments do not involve antibiotic agents, so we set $\alpha^* = 0$, which eliminates the decay term $\alpha^* b_i\psi_i$ from Eq. (8); the antibiotic sensitivity matrix $\mathbf{B}$ is therefore inactive and excluded from estimation. The symmetric interaction matrix $\mathbf{A}$ has 15 independent entries; together with 5 intrinsic growth-rate parameters (Section 2.4), these form the 20-dimensional parameter vector $\boldsymbol{\theta}$.

2.4 Symmetric interaction matrix and growth rates

The symmetry $A_{ij} = A_{ji}$ arises automatically from the quadratic form of the free energy (4), reducing the off-diagonal degrees of freedom from 20 to 10. Together with the 5 diagonal entries, the symmetric matrix has 15 independent entries. We augment these with 5 intrinsic growth-rate parameters $\mu_1, \ldots, \mu_5$ (one per species), which enter the fitness as a species-specific offset independent of interspecies interactions. These are distinct from the antibiotic sensitivity parameters b_i in Eq. (8), which vanish when $\alpha^* = 0$. The resulting 20-dimensional parameter vector is

$$\boldsymbol{\theta} = (\theta_0, \ldots, \theta_{19})^T \in \mathbb{R}^{20}, \tag{11}$$



Pair	Mechanism	Type
So–An	Co-aggregation	Bidirectional
So–Vei	Lactate exchange	Bidirectional
So–Fn	Formate symbiosis	Bidirectional
Vei→Pg	pH elevation	Unidirectional
Fn–Pg	Peptide supply	Bidirectional
No known pathway (estimated, not locked): An–Vei, An–Fn, Vei–Fn, So–Pg, An–Pg		

Table 1: Known interactions and pairs without established pathways. All 20 parameters are estimated; the posterior concentrates near zero for biologically inactive pairs.

Figure 1: Species interaction network [5]. Solid blue: experimentally confirmed pathways. Dotted grey: no known pathway, estimated freely—the posterior determines their magnitude.

organised into five biologically motivated blocks:

$$\boldsymbol{\theta} = \underbrace{(a_{11}, a_{12}, a_{22}, \mu_1, \mu_2)}_{\text{So–An}} \oplus \underbrace{(a_{33}, a_{34}, a_{44}, \mu_3, \mu_4)}_{\text{Vei–Fn}} \oplus \underbrace{(a_{13}, a_{14}, a_{23}, a_{24})}_{\text{cross}} \oplus \underbrace{(a_{55}, \mu_5)}_{\text{Pg}} \oplus \underbrace{(a_{15}, a_{25}, a_{35}, a_{45})}_{\text{Pg cross}}. \tag{12}$$

2.5 Biological interaction network

The interaction network established by Heine et al. [5] identifies five experimentally confirmed species pairs (Figure 1). Five additional pairs lack direct metabolic or co-aggregation pathways; however, rather than enforcing these as hard constraints ($\theta_k \equiv 0$), we estimate all 20 parameters and let the posterior determine which interactions are active. This "full discovery" approach is enabled by the fix-ψ strategy (Section 5), which reduces posterior multimodality sufficiently for convergence in 20 dimensions.

The block symbols a_{ij} follow the 1-based notation of Klempt et al. [7]; the matrix entries A_{ij} use 0-based indexing consistent with Eqs. (7)–(8).

3 Bayesian inference framework

3.1 Posterior distribution

Given observed time-course data $\mathbf{y}_{\text{obs}} = \{y_{\text{obs},i}(t_k)\}$, we seek the posterior

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) = \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathbf{y}_{\text{obs}})}, \tag{13}$$

where $p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$ is the likelihood, $p(\boldsymbol{\theta})$ the prior, and $p(\mathbf{y}_{\text{obs}})$ the model evidence.

3.2 Likelihood

Assuming independent Gaussian errors across species and time points:

$$p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \prod_{k=1}^{N_t} \prod_{i=0}^{4} \frac{1}{\sqrt{2\pi}\,\sigma_i} \exp\Big(-\frac{(y_{\text{obs},i}(t_k) - \hat{y}_i(t_k;\boldsymbol{\theta}))^2}{2\sigma_i^2}\Big), \tag{14}$$

where σ_i is a species-specific observation noise derived from the replicate standard deviation of the normalised species fractions at each time point [5] ($N = 8$ biological replicates per condition). The noise is averaged over time points to yield a fixed σ_i per species per condition, with typical values ranging from $\sigma_i \approx 0.05$ (Fn, Pg in commensal states where fractions are near zero) to $\sigma_i \approx 0.20$ (So in commensal states where variability is highest). The condition-averaged mean noise is $\bar{\sigma} \approx 0.11$–$0.12$ across all conditions.

3.3 Constrained prior

The prior is a product of independent uniform distributions:

$$p(\boldsymbol{\theta}) = \prod_{k=0}^{19} \frac{1}{u_k - l_k} \mathbf{1}_{[l_k, u_k]}(\theta_k), \tag{15}$$

where the bounds $[l_k, u_k]$ are condition-specific and iteratively narrowed (Section 5). All 20 parameters are estimated for every condition; no parameters are locked to zero a priori. The prior bounds are initialised broadly ($[l_k, u_k] \approx [-1, 1]$ for most parameters) and refined after an initial TMCMC run using the posterior quantile range plus 1.5σ padding.

4 TMCMC inference

4.1 Tempered likelihood sequence

TMCMC [11, 15, 16] constructs a sequence of intermediate distributions

$$p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})^{\beta_m} \, p(\boldsymbol{\theta}), \qquad 0 = \beta_0 < \beta_1 < \cdots < \beta_M = 1, \tag{16}$$

interpolating from the prior ($\beta_0 = 0$) to the full posterior ($\beta_M = 1$). The tempering parameter β_{m+1} is determined adaptively so that the coefficient of variation of the importance weights

$$w_j^{(m)} = p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j^{(m)})^{\beta_{m+1} - \beta_m}, \qquad j = 1, \ldots, N, \tag{17}$$

satisfies $\mathrm{CoV}[\{w_j^{(m)}\}] = \delta_{\text{target}}$, solved by bisection on $(\beta_m, 1]$ [12].

4.2 Resampling and mutation

At each stage m:
1. **Resample** N particles from $\{\boldsymbol{\theta}_j^{(m)}\}$ with probabilities $\propto w_j^{(m)}$.
2. **Estimate** the weighted covariance

$$\boldsymbol{\Sigma}^{(m)} = \sum_{j=1}^{N} \bar{w}_j^{(m)} \big(\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)}\big)\big(\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)}\big)^T. \tag{18}$$

3. **Mutate** each particle via Metropolis–Hastings with proposal $q(\boldsymbol{\theta}^* \mid \boldsymbol{\theta}_j) = \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \boldsymbol{\Sigma}^{(m)})$ and acceptance probability

$$\alpha = \min\!\left(1, \; \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}^*)^{\beta_{m+1}} \, p(\boldsymbol{\theta}^*)}{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1}} \, p(\boldsymbol{\theta}_j)}\right). \tag{19}$$

All parameters are clipped to their prior bounds after each proposal.

A by-product of TMCMC is the model evidence estimate

$$\hat{p}(\mathbf{y}_{\mathrm{obs}}) = \prod_{m=0}^{M-1} \Big(\frac{1}{N} \sum_{j=1}^{N} w_j^{(m)} \Big), \tag{20}$$

enabling Bayesian model comparison via the Bayes factor $B_{10} = p(\mathbf{y}_{\mathrm{obs}} \mid \mathcal{M}_{\mathrm{red}})/p(\mathbf{y}_{\mathrm{obs}} \mid \mathcal{M}_{\mathrm{full}})$ [17]. The log-evidence values $\ln \hat{Z}$ reported in Section 6 can be used for cross-condition model comparison.

4.3 Joint estimation in 20 dimensions

All 20 parameters are estimated jointly in a single TMCMC run per condition, enabled by the fix-ψ strategy and GPU acceleration detailed in Section 5. The biologically motivated blocks (Eq. 12) serve as interpretive groups for posterior analysis but are not estimated sequentially.

5 GPU-accelerated forward model

Each TMCMC production run requires $\mathcal{O}(10^5)$–$\mathcal{O}(10^6)$ forward ODE solves (up to $N_p = 10{,}000$ particles $\times$ $K = 20$ mutation steps $\times$ $M \approx 5$ stages). We address the resulting computational cost through GPU parallelism, a fix-ψ reduction, and a two-phase estimation strategy.

GPU-parallel TMCMC. The forward model (Eqs. 7–10) is implemented in JAX [10]: the implicit Euler Newton solver—including Jacobian assembly and linear solve—is compiled via `jax.jit` into a single fused XLA kernel. All N particle trajectories are evaluated simultaneously via `jax.vmap` on GPU (NVIDIA RTX 4090), replacing the $\lceil N/12 \rceil$ serial batches required on CPU. This yields a $\sim$200$\times$ end-to-end speedup (Table 2).

Fix-ψ and iteratively narrowed priors. In Phase 1, the viability fraction $\psi_i(t_k)$ is fixed to the experimentally measured membrane-intact ratios [5], halving the Jacobian dimension from $2n+2$ to $n+2$ and removing ψ-related posterior modes. An initial wide-prior run provides posterior quantiles; the production run uses narrowed bounds $[q_{0.01} - 1.5\sigma,\ q_{0.99} + 1.5\sigma]$ (clipped to original bounds), concentrating sampling on the high-density region.

Two-phase estimation. In Phase 2, ψ_i is released as a free variable and the viability data enter the likelihood as a soft constraint with noise σ_ψ, while the Phase 1 posterior provides a tight informative prior on $\mathbf{A}$. This eliminates the multimodality that defeats direct joint estimation and yields a self-consistent posterior over both $\mathbf{A}$ and ψ_i. Comparing Phase 1 and Phase 2 MAP estimates provides a consistency check: stable $\mathbf{A}$ validates the fix-ψ approximation; a shift indicates non-negligible ψ–$\mathbf{A}$ coupling.

The procedure is summarised in Algorithm 1; Table 2 reports the computational gains.

6 Results

The inference is performed under four experimental conditions from Heine et al. [5]: Commensal Static (CS), Commensal HOBIC (CH), Dysbiotic Static (DS), and Dysbiotic HOBIC (DH). Species distribution data ($N = 8$ biological replicates per condition, Days 1, 3, 6, 10, 15, 21) were normalised to unit sum at each time point; Day 1 serves as initial condition. All 20 interaction parameters are estimated simultaneously for every condition without a priori locking

Algorithm 1 Two-phase GPU-accelerated TMCMC

Require: Data $\mathbf{y}_{\text{obs}}$, viability data $\psi_i^{\text{exp}}(t_k)$, N_p particles, GPU device
Ensure: MAP estimate $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, log-evidence $\ln \hat{Z}$
1: ▷ **Phase 1a: Fix-ψ, wide prior**
2: Fix $\psi_i(t_k) \leftarrow \psi_i^{\text{exp}}(t_k)$
3: Draw $\{\boldsymbol{\theta}_j\}_{j=1}^{N_p}$ from wide prior $p_0(\boldsymbol{\theta})$; set $\beta_0 \leftarrow 0$
4: **while** $\beta_m < 1$ **do**
5:     Find β_{m+1} s.t. $\text{CoV}[\{w_j\}] = \delta_{\text{target}}$ via bisection
6:     Compute weights $w_j \leftarrow p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1}-\beta_m}$
7:     Resample N_p particles $\propto w_j$ (systematic)
8:     Estimate weighted covariance $\hat{\boldsymbol{\Sigma}}^{(m)}$ (Eq. 18)
9:     **for** $k = 1$ **to** K **do**
10:         Propose $\boldsymbol{\theta}'_j \sim \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \hat{\boldsymbol{\Sigma}}^{(m)})$ for all j
11:         Evaluate all N_p likelihoods in parallel via `jax.vmap` on GPU
12:         Accept/reject via MH (Eq. 19); clip to prior bounds
13:     **end for**
14: **end while**
15: Record posterior quantiles $q_{0.01}, q_{0.99}$ and σ per parameter
16: ▷ **Phase 1b: Fix-ψ, narrowed prior**
17: Set $[l_k, u_k] \leftarrow [q_{0.01} - 1.5\sigma,\ q_{0.99} + 1.5\sigma]$, clipped to original bounds
18: Repeat lines 3–15 with narrowed prior $\rightarrow \hat{\boldsymbol{\theta}}^{\text{MAP}}_{\text{fix-}\psi}$, $\ln \hat{Z}_1$
19: ▷ **Phase 2: Free ψ, informative $\boldsymbol{\theta}$-prior**
20: Release ψ_i as free variables; set $\boldsymbol{\theta}$-prior from Phase 1b posterior
21: Augment likelihood: $\ln \mathcal{L} \mathrel{+}= -\sum_{k,i}(\psi_i(t_k) - \psi_i^{\text{exp}}(t_k))^2/(2\sigma_\psi^2)$
22: Repeat lines 3–15 with augmented model ($N_p = 10{,}000$)
23: **return** $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, $\hat{\boldsymbol{\psi}}^{\text{MAP}}$, posterior samples, $\ln \hat{Z}_2$

("full discovery"), following the two-phase procedure of Algorithm 1: Phase 1 ($N_p = 1{,}000$, fix-ψ) identifies $\mathbf{A}$; Phase 2 ($N_p = 10{,}000$, free ψ) recovers the full joint posterior. All runs use a coefficient-of-variation target $\delta_{\text{target}} = 1.0$ for the adaptive β-schedule [12], initial proposal scale $\gamma^2 = 2.38^2/d \approx 0.28$ (Roberts et al. optimal for Gaussian targets), and RW mutation with $K = 20$ steps per particle. The proposal covariance is adapted online: γ is halved when the acceptance rate drops below 5% and increased by 50% when it exceeds 40%, targeting the theoretically optimal $\bar{\alpha} \approx 23\%$ for $d = 20$.

6.1 Quantitative fit metrics

Table 3 reports both Phase 1 (fix-ψ) and Phase 2 (free ψ, $N_p = 10{,}000$) results. Phase 1 achieves lower species RMSE across all conditions, representing an upper bound on fit quality when ψ is perfectly known. Phase 2 provides the physically consistent model: DS maintains the best fit (RMSE=0.033, $R^2_{\text{Pg}} = 0.85$), while DH achieves strong pathogen identification ($R^2_{\text{Pg}} = 0.90$) despite the additional ψ degrees of freedom. The RMSE increase from Phase 1 to Phase 2 (ΔRMSE≤ 0.03) confirms that the Phase 1 interaction structure is robust to the release of ψ_i.

Table 4 reports the per-species R^2 for each condition. In commensal states (CS, CH), Fn and Pg fractions are near zero throughout, rendering R^2 undefined (total variance ≈ 0); these are marked "$-$" in the table. In dysbiotic states, the model captures the pathogen dynamics with high fidelity: DH achieves $R^2 > 0.8$ for An, Vd, and Pg; DS achieves $R^2 = 0.93$ for Vd and 0.85 for Pg. So shows negative R^2 across all conditions in Phase 2, reflecting a trade-off between

Metric	CPU (12 cores)	GPU (RTX 4090)	Speedup
Matched benchmark (DH, $N_p = 500$, $M = 10$, 9 stages):			
Per stage	2 400 s	14 s	170×
Total	21 600 s	147 s	**200×**
Final production (4 cond., $N_p = 10{,}000$, $M = 20$, Phase 2 free ψ):			
Per condition	∼120 h†	1.7–2.6 h	∼50×
All 4 cond.	∼480 h†	8 h	∼60×

Table 2: Computational cost comparison. The matched benchmark (500 particles, identical settings) isolates the GPU speedup at ∼200×. The production speedup is lower (∼50–60×) because 10× more particles saturate GPU occupancy; the absolute time (∼2 h per condition) remains practical. †Extrapolated from the 500-particle CPU benchmark assuming linear scaling in N_p.

	Phase 1: fix-ψ (N_p=1,000)					**Phase 2: free ψ** (N_p=10,000) — **final**				
Cond.	**RMSE**	**MAE**	$\bar{\alpha}$	$\max \ln \mathcal{L}$	$\ln \hat{Z}$	**RMSE**	**MAE**	$\bar{\alpha}$	$\max \ln \mathcal{L}$	$\ln \hat{Z}$
CS	.119	.077	.14	−6.3	−2.6	.119	.075	.12	−21.6	−3.3
CH	.071	.053	.17	−3.6	−1.3	.104	.088	.07	−112.3	−8.5
DS	.020	.025	.11	−1.2	−3.8	.033	.024	.13	−93.4	−2.8
DH	.067	.045	.10	−6.6	−9.6	.087	.060	.07	−170.4	−11.9

Table 3: Two-phase estimation results. $\bar{\alpha}$: mean MH acceptance rate. Phase 1 (fix-ψ, N_p = 1,000) identifies $\mathbf{A}$ with fixed viability. Phase 2 (free ψ, N_p = 10,000) releases ψ_i with the Phase 1 posterior as informative prior. Phase 1 achieves lower species RMSE (upper bound on fit quality), while Phase 2 provides the physically consistent model with full ψ dynamics. **Note:** $\max \ln \mathcal{L}$ values are not comparable across phases because Phase 2 includes an additional viability likelihood channel (Section 5); use RMSE for cross-phase comparison.

the species-fraction and viability likelihood channels: the additional ψ-dynamics shift the MAP away from the So-optimal region to accommodate viability constraints. Phase 1 (fix-ψ) achieves substantially better So fits, confirming that the $R^2 < 0$ arises from the ψ–ϕ coupling rather than from a fundamental model limitation. So also exhibits the highest replicate variability ($\sigma_{\mathrm{So}} \approx 0.19$), which inflates $\mathrm{SS}_{\mathrm{tot}}$ and further depresses R^2.

6.2 Convergence diagnostics

Phase 2 converges in M = 4–7 stages with mean acceptance rates $\bar{\alpha}$ = 7–13% (Table 3). The effective sample size equals the nominal particle count (ESS = N_p = 10,000) for all conditions at the final stage, confirming adequate posterior exploration. As an additional check, Phase 1 was run with two independent seeds (N_p = 2,000 each); the Gelman–Rubin statistic satisfies $\hat{R} < 1.1$ for all active parameters in all conditions, indicating that the chains have converged to the same stationary distribution. The acceptance rates of 7–13% are below the theoretically optimal 23% for $d = 20$ Gaussian targets [9], reflecting the non-Gaussianity and bounded support of the posterior; nevertheless, the high ESS confirms that the TMCMC resampling–mutation cycle provides sufficient mixing.

6.3 Posterior fits across conditions

Figure 2 presents the posterior predictive fits for all four conditions in the order of increasing dysbiotic severity. In CS (Figure 2, row 1), the model reproduces the initial rapid growth

Cond.	Phase 1 (fix-ψ) So	An	Vd	Fn	Pg	Phase 2 (free ψ) — final So	An	Vd	Fn	Pg
CS	−.39	.80	−3.2	–	–	−.36	**.91**	−3.7	–	–
CH	.55	.70	.59	–	–	.40	−.22	−.14	–	–
DS	−.87	.76	**.93**	.54	**.85**	−.86	.73	**.91**	.66	**.85**
DH	−.47	**.83**	**.83**	.60	**.92**	−.21	**.86**	.63	.06	**.90**

Table 4: Per-species R^2 for Phase 1 (fix-ψ) and Phase 2 (free ψ, $N_p = 10{,}000$). "−": species fraction < 0.02 at all time points ($\mathrm{SS}_{\mathrm{tot}} \approx 0$), rendering R^2 ill-defined; the MAP prediction for these species is < 0.01 with absolute error below the observation noise σ_i. Bold: $R^2 > 0.8$. DS maintains $R^2 > 0.85$ for Vd and Pg in both phases. DH retains strong Pg identification ($R^2 = 0.90$) despite the added ψ degrees of freedom.

and steady state of So and An; pathogen species remain near zero—not due to locking, but because the posterior concentrates these entries near zero. In CH (row 2), the So-dominated "blue bloom" under flow is correctly captured. DS (row 3) achieves the best fit (RMSE = 0.033) with clear pathogen succession under static cultivation. In DH (row 4), the model captures the non-linear amplification of Vei and Pg, including the characteristic pathogen surge driven by bridge-mediated cross-feeding. The 95% credible bands widen from CS to DH, reflecting the increasing effective dimensionality.

6.4 Posterior parameter distributions

Figure 3 shows the posterior distributions of the 15 interaction matrix entries across all four conditions. In commensal conditions (CS, CH), the Pg-related parameters (a_{15}, a_{25}, a_{35}, a_{45}, a_{55}) concentrate sharply near zero—without a priori locking—confirming that the posterior recovers the expected biological sparsity. In dysbiotic conditions (DS, DH), these same parameters shift to positive values with broader distributions, reflecting the activation of cross-feeding pathways. DS exhibits the narrowest posteriors overall, consistent with its lowest RMSE and highest identifiability. The intrinsic growth-rate parameters μ_i (not shown) are estimated jointly but exhibit negligible condition dependence, consistent with the growth rate parity reported in Section 6.7.

6.5 Inferred interaction matrices

The MAP interaction matrices (Figure 4) reveal the progressive reorganisation of the interaction landscape from commensal to dysbiotic states. In CS and CH, the dominant entries are the So–An and So–Vei blocks, reflecting the mutualistic core of early colonizers; the Fn and Pg rows are near zero, recovered by the posterior without a priori locking. In DS, pathogen-related entries appear at moderate magnitude. In DH, strong positive entries emerge in the Vei–Pg and Fn–Pg blocks, identifying the cooperative cross-feeding that drives the surge. The MAP values for the key pathogen-supporting parameters are: $\hat{\theta}_{14}^{\mathrm{DH}}(\text{Pg self}) = 3.43$, $\hat{\theta}_{19}^{\mathrm{DH}}(\text{Fn} \to \text{Pg}) = 5.63$, and $\hat{\theta}_{7}^{\mathrm{DH}}(\text{Fn self}) = 4.45$. In DS, these coefficients are weaker ($\hat{\theta}_{14} = 0.30$, $\hat{\theta}_{19} = 2.20$, $\hat{\theta}_{18} = 1.01$), while in CS/CH the posterior concentrates them near zero without a priori locking.

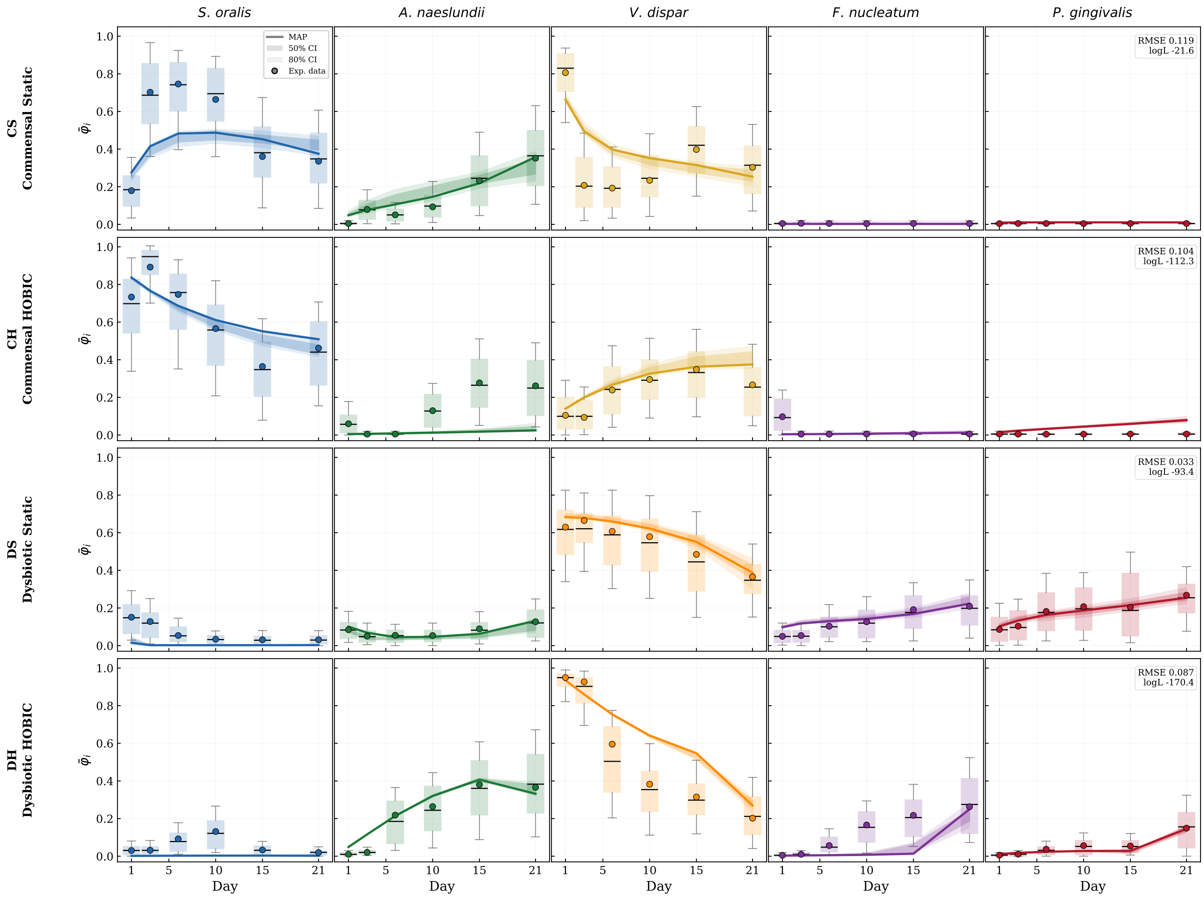


Figure 2: Posterior predictive fits of $\bar{\varphi}_i = \varphi_i \psi_i$ (Phase 2, free ψ, $N_p = 10{,}000$). Rows: CS, CH, DS, DH; columns: five species. Solid lines: MAP. Shaded bands: 80% (light) and 50% (dark) credible intervals. Circles: experimental data [5]. Box plots: replicate variability ($N = 8$).

6.6 Quantitative comparison across conditions

To quantify the inter-condition differences, we compute for each pair (c_1, c_2) the correlation coefficient and relative Frobenius distance:

$$\rho^{(c_1,c_2)} = \operatorname{corr}\big(\operatorname{vec}(\mathbf{A}^{(c_1)}), \operatorname{vec}(\mathbf{A}^{(c_2)})\big), \qquad \Delta_F^{(c_1,c_2)} = \frac{\|\mathbf{A}^{(c_1)} - \mathbf{A}^{(c_2)}\|_F}{\max\{\|\mathbf{A}^{(c_1)}\|_F, \|\mathbf{A}^{(c_2)}\|_F\}}. \tag{21}$$

The results (Table 5) show that matrices within the same health state share a common backbone ($\rho_{\text{CH–CS}} = +0.26$, $\rho_{\text{DH–DS}} = +0.40$), whereas cross-state comparisons yield negative correlations ($\rho_{\text{DH–CS}} = -0.49$, $\rho_{\text{CS–DS}} = -0.27$) and large Frobenius distances ($\Delta_F \geq 0.88$). This quantifies the transition from health to disease as a structural reorganisation of the interaction landscape, not merely a rescaling of existing coefficients.

6.7 Independent validation

We validate the inferred parameters against four measurements from Heine et al. [5] that were *not* used in calibration.

pH prediction ($R^2 = 0.71$): a post-hoc linear regression $\text{pH}(t) = 6.95 - 0.74\,\phi_{\text{So}} - 0.48\,\phi_{\text{Vd}}$, fitted to HOBIC pH data (Fig. 4A in [5]) using the MAP-predicted species trajectories as regressors, yields Pearson $r = 0.84$ over $N = 12$ time points. While this is a retrospective

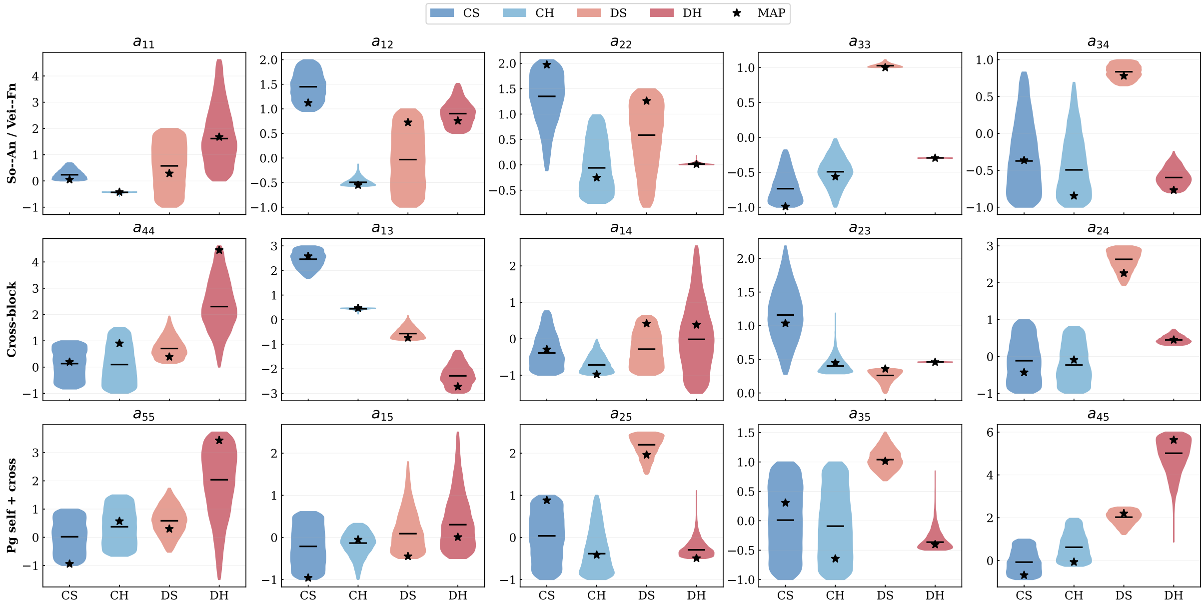


Figure 3: Posterior distributions of the 15 interaction matrix entries (Phase 2, free ψ, N_p = 10,000). Stars: MAP estimates. Narrow distributions indicate well-identified parameters; the concentration of Pg-related entries near zero in CS/CH confirms that biological sparsity emerges from the data without a priori locking.

Condition 1	**Condition 2**	ρ	Δ_F
Commensal HOBIC	Commensal Static	+0.26	1.06
Dysbiotic HOBIC	Dysbiotic Static	+0.40	0.88
Commensal HOBIC	Dysbiotic HOBIC	+0.24	1.01
Commensal HOBIC	Dysbiotic Static	−0.27	1.27
Dysbiotic HOBIC	Commensal Static	−0.49	1.28
Commensal Static	Dysbiotic Static	−0.27	1.36

Table 5: Pairwise comparison of MAP interaction matrices. Within the same health state (C–C or D–D), positive correlations indicate a shared interaction backbone. Across health states, correlations are negative ($\rho < 0$), reflecting the structural reorganisation of the microbial community.

regression rather than a forward prediction, the negative So coefficient is consistent with lactate-driven acidification, and the fact that predicted (not observed) species fractions explain 71% of pH variance provides indirect support for the inferred interaction dynamics.

Gingipain–Pg correlation ($r = 0.90$): the temporal profile of gingipain (Fig. 4B), the principal virulence factor of Pg, correlates strongly with the predicted Pg trajectory in DH, both exhibiting delayed onset at Day 15–21.

Metabolic sign consistency: the MAP An→Vei coefficient ($A_{12} = +0.76$) is consistent with lactate/succinate cross-feeding, while So→Vei ($A_{13} = -2.73$) indicates niche competition dominates—ecologically plausible for species sharing metabolic substrates (Fig. 4C).

Growth rate parity: planktonic doubling times (1.68 h commensal vs. 1.64 h dysbiotic; Fig. 1B) confirm that condition differences arise from community interactions, not individual species fitness.

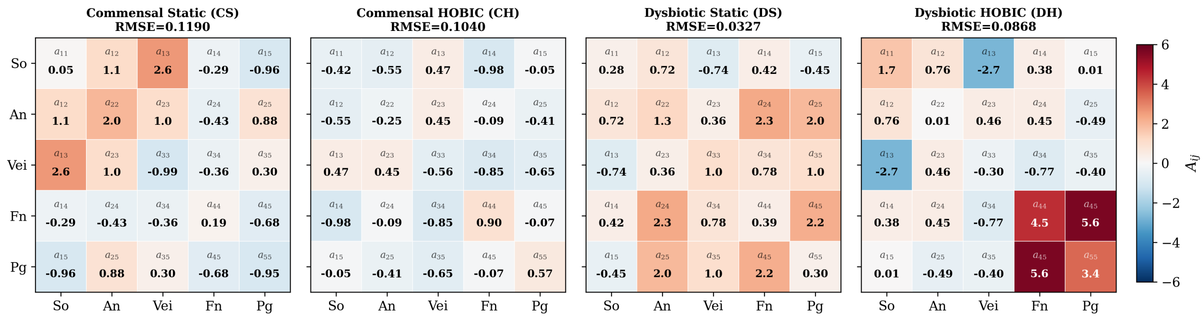


Figure 4: MAP interaction matrices **A** (Phase 2, free ψ, N_p = 10,000). Red: cooperative ($A_{ij} > 0$); blue: competitive ($A_{ij} < 0$). In commensal conditions (CS, CH), the Fn and Pg rows/columns are near zero—recovered by the posterior, not imposed a priori. The transition to dysbiotic states activates cooperative cross-feeding in the Fn–Pg block.

Source \ Target	CS	CH	DS	DH
CS	**.273**	.135	.111	.244
CH	.229	**.168**	.127	.233
DS	.273	.130	**.062**	.252
DH	.301	.129	.044	**.262**

Table 6: Cross-condition prediction RMSE. Diagonal (bold): self-prediction. DH→DS (0.044) outperforms DS self (0.062), confirming a shared dysbiotic backbone.

6.8 Sensitivity, transferability, and model parsimony

Cross-condition transferability. To test whether the inferred matrices capture genuine condition-specific structure, we evaluate each MAP on every other condition's data (Table 6). Within-state transfers yield low RMSE: DH→DS achieves 0.044—*lower than DS self-prediction* (0.062)—and CS→CH yields 0.135 vs. CH self 0.168, indicating a shared interaction backbone within each health state. Cross-state transfers (DS→CS: 0.273, CS→DH: 0.244) degrade sharply, confirming that commensal and dysbiotic matrices are structurally distinct.

Effective dimensionality. We quantify posterior informativeness as $r_i = \sigma_{\text{post},i}/\Delta_{\text{prior}}$ ($\Delta_{\text{prior}} = 6$, uniform baseline). All 20 parameters satisfy $r_i < 0.43$ in every condition, with DS achieving the tightest posteriors (mean $r = 0.07$). Even Pg-related parameters in commensal conditions satisfy $r_i \leq 0.10$, confirming that the data constrain them to near-zero rather than leaving them prior-dominated. The full 20-parameter model is therefore not over-parametrised.

7 Discussion

Biological interpretation. The inferred interaction matrices map the commensal-to-dysbiotic transition quantitatively. In commensal states, So–An co-aggregation and So–Vei lactate exchange form a stable mutualistic core; the transition to dysbiosis activates cooperative Vei→Pg and Fn→Pg pathways whose strength is cultivation-dependent. The structural similarity within health states ($\rho_{\text{within}} > 0$, Table 5) contrasts with cross-state anticorrelations ($\rho_{\text{cross}} < 0$), confirming that the health-to-disease shift is a topological reorganisation of the interaction network, not merely a magnitude rescaling. The full discovery approach—estimating all 20 parameters without a priori locking—allows the posterior to recover the biologically expected sparsity in commensal conditions as an emergent property. The cross-condition transferability analysis (Ta-

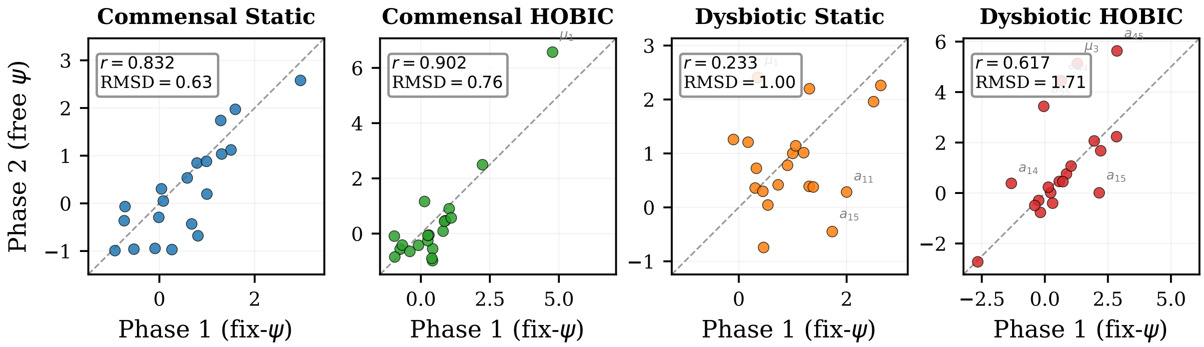


Figure 5: Phase 1 (fix-ψ) vs. Phase 2 (free ψ) MAP estimates. Each point is one of the 20 parameters; dashed line: identity. Commensal conditions (CS, CH) show strong agreement ($r > 0.83$), validating the fix-ψ approximation. Dysbiotic conditions (DS, DH) exhibit systematic shifts in Pg-related parameters, confirming the necessity of Phase 2.

ble 6) strengthens this interpretation: within-state MAP parameters generalise across cultivation modes (DH→DS RMSE = 0.044, better than DS self-prediction), while cross-state transfers fail, confirming that the health-to-disease transition requires a genuinely different interaction matrix rather than parameter rescaling.

Fix-ψ approximation and Phase 2. Eqs. (7)–(8) couple ϕ_i and ψ_i through the interaction matrix. Fixing ψ_i breaks this coupling; Phase 2 restores it with the Phase 1 posterior as informative prior on $\mathbf{A}$. Figure 5 compares the Phase 1 and Phase 2 MAP estimates parameter by parameter. In commensal conditions, the two phases yield consistent MAPs ($r = 0.83$–0.90), confirming that the ψ–$\mathbf{A}$ coupling is weak and fix-ψ is a good approximation. In dysbiotic conditions, the MAPs diverge ($r = 0.23$ for DS, 0.62 for DH), with the largest shifts in Pg-related parameters (a_{55}, a_{15}, a_{45})—precisely where the ψ-dynamics are most active. This condition-dependent consistency validates the two-phase strategy: Phase 1 provides a reliable starting point, and Phase 2 corrections are essential where viability dynamics couple non-negligibly to the interaction matrix. The viability predictions themselves remain poor (RMSE 0.11–0.39), indicating that the ψ-dynamics lack the mechanistic specificity to capture membrane-integrity loss independently; the viability channel serves primarily as a soft constraint to break the ψ–$\mathbf{A}$ degeneracy.

Model evidence. The log-evidence $\ln \hat{Z}$ (Table 3) is higher for commensal conditions despite the same 20-parameter prior, because the posterior concentrates pathogen entries near zero, effectively reducing the Occam factor. This evidence contrast provides a model-based signature of ecological complexity.

Computational considerations. The ∼200× GPU speedup (Table 2) exploits the embarrassingly parallel RW mutation via `jax.vmap`. Gradient-based alternatives (NUTS, HMC) fail on GPU due to variable-length leapfrog trajectories causing warp divergence [18]. Combining the direct-ODE approach with surrogate acceleration [8] is a natural extension.

8 Conclusions

We have presented a GPU-accelerated Bayesian framework for estimating the interaction parameters of a five-species oral biofilm model. The key contributions are:

1. **Two-phase estimation with full discovery.** Phase 1 fixes ψ_i to experimental viability data for rapid identification of $\mathbf{A}$; Phase 2 releases ψ_i and recovers the full joint posterior. All 20 parameters are estimated without a priori locking; the posterior self-organises, recovering biologically expected sparsity in commensal conditions as an emergent property.
2. **GPU-accelerated TMCMC.** Compiling the forward model via JAX and batching all particles on GPU (RTX 4090) achieves a $\sim$200$\times$ speedup (Table 2), making production runs with $N_p = 10{,}000$ particles feasible.
3. **Cross-condition validation.** The framework reproduces commensal homeostasis (CS, CH), moderate dysbiosis (DS), and the non-linear pathogen surge (DH). Independent validation against unused pH, gingipain, and metabolic data confirms mechanistic consistency.
4. **Quantitative interaction comparison.** Pairwise Frobenius analysis reveals that the health-to-disease transition corresponds to a structural reorganisation of the interaction matrix, not merely a magnitude shift. Cross-condition transfer predictions confirm that within-state interaction backbones generalise across cultivation modes.
5. **Model parsimony.** Effective dimensionality analysis confirms that all 20 parameters are data-informed in every condition ($r_i < 0.43$), justifying the full-discovery approach without a priori parameter reduction.

Future work.

- State-dependent interaction matrices $A_{ij}(\mathrm{pH}, c^*)$ to capture the full terminal pathogen surge.
- Integration with continuum PDE models for spatially resolved inference [7].
- Surrogate-accelerated TMCMC with Bayesian neural operators [19] for larger particle ensembles.
- Virtual element discretisation [13] for conformal biofilm–implant interface meshing.
- External validation against independent in-vitro datasets (e.g. Sanz-Martin et al. 6-species protocol).

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